1. Print your current working directory to verify your location in the filesystem.

```
akansha@localhost:~$ pwd
/home/akansha
```

2. List all files and directories in your current location using the basic ls command.

```
akansha@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

3. List all contents of your home directory, including hidden files (starting with a dot), using appropriate flags.

```
akansha@localhost:~$ ls -a
.   ..  .bash_history  .bash_logout  .bash_profile  .bashrc  .cache  .config
Desktop  Documents  Downloads
```

4. Display all files in long-listing format with human-readable file sizes using ls.

```
akansha@localhost:~$ ls -lh
total 32K
drwxr-xr-x. 2 akansha akansha 4.0K Oct 31 10:00 Desktop
drwxr-xr-x. 2 akansha akansha 4.0K Oct 31 10:00 Documents
drwxr-xr-x. 2 akansha akansha 4.0K Oct 31 10:00 Downloads
```

5. Create a new directory named myfirstlab in your current directory.

```
akansha@localhost:~$ mkdir myfirstlab
akansha@localhost:~$ ls -d myfirstlab
myfirstlab
```

6. Navigate into the newly created myfirstlab directory using cd.

```
akansha@localhost:~$ cd myfirstlab
akansha@localhost:~/myfirstlab$ pwd
/home/akansha/myfirstlab
```

7. Inside myfirstlab, create three empty text files: a.txt, b.txt, and c.txt using a single command.

```
akansha@localhost:~/myfirstlab$ touch a.txt b.txt c.txt
akansha@localhost:~/myfirstlab$ ls -l
total 0
-rw-r--r--. 1 akansha akansha 0 Oct 31 10:05 a.txt
-rw-r--r--. 1 akansha akansha 0 Oct 31 10:05 b.txt
-rw-r--r--. 1 akansha akansha 0 Oct 31 10:05 c.txt
```

8. Display the full contents of the system file /etc/os-release on your terminal screen.

```
akansha@localhost:~/myfirstlab$ cat /etc/os-release
NAME="Red Hat Enterprise Linux"
VERSION="8.10 (Ootpa)"
ID="rhel"
ID_LIKE="fedora"
VERSION_ID="8.10"
PLATFORM_ID="platform:el8"
PRETTY_NAME="Red Hat Enterprise Linux 8.10 (Ootpa)"
ANSI_COLOR="0;31"
CPE_NAME="cpe:/o:redhat:enterprise_linux:8::baseos"
HOME_URL="https://www.redhat.com/"
```

9. Check the username of the user currently logged into the terminal session.

```
akansha@localhost:~/myfirstlab$ whoami
akansha
```

10. Display your system's hostname on the terminal.

```
akansha@localhost:~/myfirstlab$ hostname
localhost.localdomain
```

11. Check how much disk space is currently free on your system in human-readable format.

```
akansha@localhost:~/myfirstlab$ df -h
Filesystem             Size  Used Avail Use% Mounted on
devtmpfs               1.9G     0  1.9G   0% /dev
/dev/mapper/rhel-root   27G  4.2G   23G  16% /
/dev/sda1             1014M  240M  775M  24% /boot
```

12. Check the total system RAM and current memory usage in human-readable units.

```
akansha@localhost:~/myfirstlab$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       1.1Gi       1.4Gi        12Mi       1.2Gi       2.3Gi
Swap:          2.0Gi          0B       2.0Gi
```

13. Check how long your Linux system has been continuously running.

```
akansha@localhost:~/myfirstlab$ uptime
 10:15:30 up  3:45,  1 user,  load average: 0.04, 0.03, 0.00
```

14. Display full system and kernel operating information using uname.

```
akansha@localhost:~/myfirstlab$ uname -a
Linux localhost.localdomain 4.18.0-553.el8_10.x86_64 #1 SMP Wed May 1 09:54:27
EDT 2026 x86_64 x86_64 x86_64 GNU/Linux
```

15. Use the manual pages to look up the documentation for the ls command.

```
akansha@localhost:~/myfirstlab$ man ls
# Press 'q' to quit the manual page viewer.
```

16. Display a short quick-usage summary for the cp command using the --help flag.

```
akansha@localhost:~/myfirstlab$ cp --help
Usage: cp [OPTION]... [-T] SOURCE DEST
  or:  cp [OPTION]... SOURCE... DIRECTORY
  or:  cp [OPTION]... -t DIRECTORY SOURCE...
Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.
```

17. Use the whatis command to get a one-line description of the mv command.

```
akansha@localhost:~/myfirstlab$ whatis mv
mv (1)               - move (rename) files
```

18. Navigate back to your home directory using the shorthand tilde (~) symbol.

```
akansha@localhost:~/myfirstlab$ cd ~
akansha@localhost:~$ pwd
/home/akansha
```

19. Create a nested folder hierarchy project/src/data in a single command using mkdir.

```
akansha@localhost:~$ mkdir -p project/src/data
akansha@localhost:~$ ls -R project
project:
src

project/src:
data

project/src/data:
```

20. Delete the file c.txt you created inside the myfirstlab directory.

```
akansha@localhost:~$ rm myfirstlab/c.txt
akansha@localhost:~$ ls myfirstlab
a.txt  b.txt
```

21. Create a file named students.txt and add 5 student names (one per line) using echo and the append redirection operator (>>).

```
akansha@localhost:~$ echo "Aarav Sharma" >> students.txt
akansha@localhost:~$ echo "Ananya Verma" >> students.txt
akansha@localhost:~$ echo "Rohan Gupta" >> students.txt
akansha@localhost:~$ echo "Riya Sen" >> students.txt
akansha@localhost:~$ echo "Vikram Malhotra" >> students.txt
akansha@localhost:~$ cat students.txt
Aarav Sharma
Ananya Verma
Rohan Gupta
Riya Sen
Vikram Malhotra
```

22. Display only the first 2 names from students.txt using the head command.

```
akansha@localhost:~$ head -n 2 students.txt
Aarav Sharma
Ananya Verma
```

23. Display only the last 2 names from students.txt using the tail command.

```
akansha@localhost:~$ tail -n 2 students.txt
Riya Sen
Vikram Malhotra
```

24. Create a directory named backup and copy students.txt into it without navigating out of your current directory.

```
akansha@localhost:~$ mkdir backup
akansha@localhost:~$ cp students.txt backup/
akansha@localhost:~$ ls backup
students.txt
```

25. Rename students.txt to class_list.txt using the mv command.

```
akansha@localhost:~$ mv students.txt class_list.txt
akansha@localhost:~$ ls class_list.txt
class_list.txt
```

26. Use grep to search inside class_list.txt for any name containing a specific letter (e.g., 'a' or 'R').

```
akansha@localhost:~$ grep "R" class_list.txt
Rohan Gupta
Riya Sen
```

27. Display line numbers along with matching text when searching for a string in class_list.txt using grep -n.

```
akansha@localhost:~$ grep -n "R" class_list.txt
3:Rohan Gupta
4:Riya Sen
```

28. Change permissions of class_list.txt so only the owner has read and write rights (no execute), and verify using long listing.

```
akansha@localhost:~$ chmod 600 class_list.txt
akansha@localhost:~$ ls -l class_list.txt
-rw-------. 1 akansha akansha 65 Oct 31 10:15 class_list.txt
```

29. Find all files ending with .txt in your home directory tree using the find command.

```
akansha@localhost:~$ find ~ -name "*.txt"
/home/akansha/class_list.txt
/home/akansha/myfirstlab/a.txt
/home/akansha/myfirstlab/b.txt
/home/akansha/backup/students.txt
```

30. Use wc -l to count the exact number of lines (students) in class_list.txt.

```
akansha@localhost:~$ wc -l class_list.txt
5 class_list.txt
```

31. Display the first 5 lines of /etc/passwd on the terminal.

```
akansha@localhost:~$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
```

32. Display the last 5 lines of /var/log/dpkg.log or /var/log/syslog.

```
akansha@localhost:~$ sudo tail -n 5 /var/log/messages
Oct 31 08:30:02 localhost systemd[1]: Started Network Manager.
Oct 31 08:30:05 localhost systemd[1]: Reached target Multi-User System.
Oct 31 10:00:12 localhost dnf[1245]: Metadata cache refreshed.
Oct 31 10:15:00 localhost sshd[3238]: Accepted password for akansha.
Oct 31 10:15:01 localhost systemd[1]: Started Session 1 of user akansha.
```

33. Copy an entire directory structure (project/) into a new folder named project_backup/ recursively.

```
akansha@localhost:~$ cp -r project/ project_backup/
akansha@localhost:~$ ls -R project_backup
project_backup:
src

project_backup/src:
data
```

34. View a large text file page-by-page using less and identify the key used to quit the pager.

```
akansha@localhost:~$ less /etc/services
# Press 'q' to quit the pager.
```

35. Move two files (a.txt and b.txt) simultaneously into a target folder named archive/.

```
akansha@localhost:~$ mkdir archive
akansha@localhost:~$ mv myfirstlab/a.txt myfirstlab/b.txt archive/
akansha@localhost:~$ ls archive
a.txt  b.txt
```

36. Combine commands: Create a directory logs, inside it create app.log with 10 lines of dummy text using a loop or repeated echo >>, then view the last 3 lines using tail.

```
akansha@localhost:~$ mkdir logs && for i in {1..10}; do echo "Log entry line $i"
>> logs/app.log; done && tail -n 3 logs/app.log
Log entry line 8
Log entry line 9
Log entry line 10
```

37. Use find paired with -exec to locate all .log files in your current working path and display their contents in a single command line.

```
akansha@localhost:~$ find . -name "*.log" -exec cat {} +
Log entry line 1
Log entry line 2
Log entry line 3
Log entry line 4
Log entry line 5
Log entry line 6
Log entry line 7
Log entry line 8
Log entry line 9
Log entry line 10
```

38. Recursively search all files in your current directory tree for the word 'error' using grep -r.

```
akansha@localhost:~$ echo "An error occurred" >> logs/app.log
akansha@localhost:~$ grep -r "error" .
./logs/app.log:An error occurred
```

39. Create a shell script test.sh, grant it execute permissions using symbolic chmod syntax (+x), and write a comment inside explaining the permission string rwxr-xr--.

```
akansha@localhost:~$ cat << 'EOF' > test.sh
#!/bin/bash
# Permission rwxr-xr-- Explanation:
# Owner (u)  : rwx (read, write, execute)
# Group (g)  : r-x (read, execute, no write)
# Others (o) : r-- (read only, no write/execute)
echo "Executing test script"
EOF
akansha@localhost:~$ chmod +x test.sh
akansha@localhost:~$ ls -l test.sh
-rwxr-xr-x. 1 akansha akansha 215 Oct 31 10:25 test.sh
```

40. Write a single-line command using chaining (&&) to create a folder demo, enter it, and list its contents.

```
akansha@localhost:~$ mkdir demo && cd demo && ls -la
total 12
drwxr-xr-x. 2 akansha akansha 4096 Oct 31 10:30 .
drwxr-xr-x. 8 akansha akansha 4096 Oct 31 10:30 ..
```

41. Combine find and wc -l in a single pipeline to count how many total .txt files exist under your home directory tree.

```
akansha@localhost:~$ find ~ -type f -name "*.txt" | wc -l
4
```

42. Find all directories (not regular files) inside /var/log using find -type d.

```
akansha@localhost:~$ find /var/log -type d
/var/log
/var/log/audit
/var/log/sssd
/var/log/tuned
/var/log/chrony
```

43. Search for a specific file system configuration entry inside /etc/fstab using grep, ignoring case sensitivity.

```
akansha@localhost:~$ grep -i "xfs" /etc/fstab
/dev/mapper/rhel-root   /                       xfs     defaults        0 0
UUID=a1b2c3d4-e5f6-7890-abcd-111111111111 /boot  xfs     defaults        0 0
```

44. Check the disk space consumption of a specific folder in human-readable format using du -sh.

```
akansha@localhost:~$ du -sh logs
8.0K    logs
```

45. Write a command that displays the 3 largest files in a directory by sorting ls output by file size.

```
akansha@localhost:~$ ls -lS /var/log | head -n 4
total 12540
-rw-------. 1 root root 5242880 Oct 31 10:00 messages
-rw-------. 1 root root 3145728 Oct 31 09:30 secure
-rw-------. 1 root root 1048576 Oct 31 08:00 maillog
```

46. Debug / Safety Habit: Explain why running rm -rf * in an unverified path is hazardous, and state the safety verification command habit that prevents accidental deletion.

```
akansha@localhost:~$ pwd
/home/akansha/demo

Explanation & Safety Habit:
- Hazard: 'rm -rf *' recursively and forcefully deletes all files and
subdirectories without confirmation. If executed in root (/) or home directory
due to an undetected 'cd' failure, it causes unrecoverable system loss.
- Safety Habit: ALWAYS verify location using 'pwd' and test directory contents
with 'ls' before running any deletion command. Alternatively, use 'rm -i' for
interactive prompt verification.
```

47. Use touch on an existing file to update its access and modification timestamp to the current system time without modifying its contents.

```
akansha@localhost:~$ ls -l class_list.txt
-rw-------. 1 akansha akansha 65 Oct 31 10:15 class_list.txt
akansha@localhost:~$ touch class_list.txt
akansha@localhost:~$ ls -l class_list.txt
-rw-------. 1 akansha akansha 65 Oct 31 10:35 class_list.txt
```

48. Create a file containing multiple duplicate lines, then use sort and uniq to display only unique lines from the file.

```
akansha@localhost:~$ echo -e "apple
banana
apple
orange
banana" > dup.txt
akansha@localhost:~$ sort dup.txt | uniq
apple
banana
orange
```

49. Redirect the stdout of a command (e.g., uname -a) to a file named sys_info.txt, then view its content.

```
akansha@localhost:~$ uname -a > sys_info.txt
akansha@localhost:~$ cat sys_info.txt
Linux localhost.localdomain 4.18.0-553.el8_10.x86_64 #1 SMP Wed May 1 09:54:27
EDT 2026 x86_64 x86_64 x86_64 GNU/Linux
```

50. Find all files in /var/log that were modified within the last 7 days using find -mtime.

```
akansha@localhost:~$ find /var/log -type f -mtime -7
/var/log/messages
/var/log/secure
/var/log/maillog
/var/log/cron
```